

L2 : MORE ABOUT FACTORIZATION (B)

Checkpoint 1

$$a^2 + 2ab + b^2 = (a + b)^2$$

Checkpoint 2

$$a^2 - b^2 = (a + b)(a - b)$$

Checkpoint 3

Factorize $x^2 + 3x + 2$

x^2		2
x	+2	
x	+1	



 $x + 2x = 3x$

 $(x)(+1) + (x)(+2) = 3x$

 $x^2 + 3x + 2$

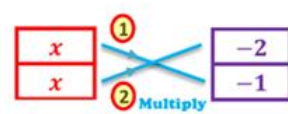
 $\Rightarrow \begin{matrix} x & +2 \\ x & +1 \end{matrix} \Rightarrow (x+2)(x+1)$

 $x^2 + 3x + 2 = (x+2)(x+1)$

Checkpoint 4

Factorize $x^2 - 3x + 2$

x^2		2
x	+2	-2
x	+1	-1



 $(-x) + (-2x) = -3x$

 $(x)(-1) + (x)(-2) = -3x$

 $x^2 - 3x + 2$

 $\Rightarrow \begin{matrix} x & -2 \\ x & -1 \end{matrix} \Rightarrow (x-2)(x-1)$

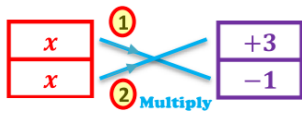
 $x^2 - 3x + 2 = (x-2)(x-1)$

Checkpoint 5

Factorize $x^2 + 2x - 3$

Trial 1

x^2		-3
x	+3	-3
x	-1	+1



 $(-x) + (3x) = 2x$

 $(x)(-1) + (x)(+3) = 2x$

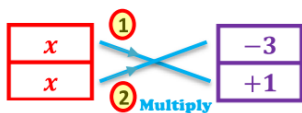
 $x^2 + 2x - 3$

最尾係正，正正負負

最尾係負，正負正負

Trial 2

x^2		-3
x	+3	-3
x	-1	+1



 $(x) + (-3x) = -2x$

 $(x)(+1) + (x)(-3) = -2x$

 $x^2 + 2x - 3$

Final Ans:

$$x^2 + 2x - 3 = (x + 3)(x - 1)$$

PRE S3 MATHS L2 – MORE ABOUT FACTORIZATION (B)

Let's get started!

Factorize the following expressions.

1. $a^2 + 24a + 95$

$$\underline{\underline{(a+5)(a+19)}}$$

2. $b^2 - 48b + 176$

$$\underline{\underline{(b-4)(b-44)}}$$

3. $c^2 - 13c - 90$

$$\underline{\underline{(c+5)(c-18)}}$$

4. $d^2 + 17d - 200$

$$\underline{\underline{(d-8)(d+25)}}$$

5. $66 + w^2 + 17w$

$$\underline{\underline{(w+6)(w+11)}}$$

6. $-6s + s^2 - 91$

$$\underline{\underline{(s+7)(s-13)}}$$

7. $-32 - 4h + h^2$

$$\underline{\underline{(h+4)(h-8)}}$$

8. $-42 + 11r + r^2$

$$\underline{\underline{(r-3)(r+14)}}$$

9. $15p - p^2 - 44$

$$\begin{aligned} 15p - p^2 - 44 &= -p^2 + 15p - 44 \\ &= -(p^2 - 15p + 44) \\ &= \underline{\underline{-(p-4)(p-11)}} \end{aligned}$$

10. $-17n - n^2 - 72$

$$\begin{aligned} -17n - n^2 - 72 &= -n^2 - 17n - 72 \\ &= -(n^2 + 17n + 72) \\ &= \underline{\underline{-(n+8)(n+9)}} \end{aligned}$$

11. $2s^2 - 26s + 72$

$$\begin{aligned} 2s^2 - 26s + 72 &= 2(s^2 - 13s + 36) \\ &= \underline{\underline{2(s-4)(s-9)}} \end{aligned}$$

12. $3t^2 - 12t - 63$

$$\begin{aligned} 3t^2 - 12t - 63 &= 3(t^2 - 4t - 21) \\ &= \underline{\underline{3(t+3)(t-7)}} \end{aligned}$$

PRE S3 MATHS L2 - MORE ABOUT FACTORIZATION (B)

Factorize the following expressions.

13. $7x^2 - 10x + 3$

$$\underline{\underline{(x-1)(7x-3)}}$$

14. $5y^2 - 11y + 2$

$$\underline{\underline{(y-2)(5y-1)}}$$

15. $4c^2 + 9c + 2$

$$\underline{\underline{(c+2)(4c+1)}}$$

16. $4s^2 - 11s + 6$

$$\underline{\underline{(s-2)(4s-3)}}$$

17. $2a^2 - 5ab + 2b^2$

$$\underline{\underline{(a-2b)(2a-b)}}$$

18. $3p^2 - 8pq + 5q^2$

$$\underline{\underline{(p-q)(3p-5q)}}$$

19. $5f^2 + 24fg + 16g^2$

$$\underline{\underline{(f+4g)(5f+4g)}}$$

20. $6s^2 + 17st + 12t^2$

$$(2s+3t)(3s+4t)$$

21. (a) $2s^2 - 11s + 14$

(b) $(s-2)^2 + 2s^2 - 11s + 14$

(a) $2s^2 - 11s + 14 = \underline{\underline{(s-2)(2s-7)}}$

(b) $(s-2)^2 + 2s^2 - 11s + 14 = (s-2)^2 + (s-2)(2s-7)$ (from (a))
 $= (s-2)[(s-2) + (2s-7)]$
 $= (s-2)(3s-9)$
 $= (s-2)3(s-3)$
 $= \underline{\underline{3(s-2)(s-3)}}$

22. (a) $3a^2 - 7ac + 2c^2$

(b) $3a^2 - 7ac + 2c^2 - ab + 2bc$

(a) $3a^2 - 7ac + 2c^2 = \underline{\underline{(a-2c)(3a-c)}}$

(b) $3a^2 - 7ac + 2c^2 - ab + 2bc$
 $= (a-2c)(3a-c) - ab + 2bc$ (from (a))
 $= (a-2c)(3a-c) - b(a-2c)$
 $= \underline{\underline{(a-2c)(3a-c-b)}}$

23. (a) $7m^2 + 16mn + 4n^2$

(b) $7m^2 + 16mn + 4n^2 + 5m + 10n$

(a) $7m^2 + 16mn + 4n^2 = \underline{\underline{(m+2n)(7m+2n)}}$

(b) $7m^2 + 16mn + 4n^2 + 5m + 10n$
 $= (m+2n)(7m+2n) + 5(m+2n)$
 $= \underline{\underline{(m+2n)(7m+2n+5)}}$

24. (a) $x^2 - 8x + 12$

(b) $xy - x^2 - 6y + 8x - 12$

(a) $x^2 - 8x + 12 = \underline{\underline{(x-2)(x-6)}}$

(b) $xy - x^2 - 6y + 8x - 12$
 $= xy - 6y - (x^2 - 8x + 12)$
 $= y(x-6) - (x-2)(x-6)$
 $= (x-6)[y - (x-2)]$
 $= \underline{\underline{(x-6)(y-x+2)}}$

PRE S3 MATHS L2 – MORE ABOUT FACTORIZATION (B)

Factorize the following expressions.

25. $4a^2 + 16a - 9$

$$\underline{\underline{(2a - 1)(2a + 9)}}$$

26. $24b^2 - b - 10$

$$\underline{\underline{(3b - 2)(8b + 5)}}$$

27. $22c^2 - 67c + 30$

$$\underline{\underline{(2c - 5)(11c - 6)}}$$

28. $9d^2 - 4d - 5$

$$\underline{\underline{(d - 1)(9d + 5)}}$$

29. $21a^2 - 8ab - 4b^2$

$$\underline{\underline{(3a - 2b)(7a + 2b)}}$$

30. $4m^2 + 4mn - 15n^2$

$$\underline{\underline{(2m - 3n)(2m + 5n)}}$$

31. $10p^2 + 9pq - 7q^2$

$$\underline{\underline{(2p - q)(5p + 7q)}}$$

32. $12u^2 - 7uv - 12v^2$

$$\underline{\underline{(3u - 4v)(4u + 3v)}}$$

33. $7m^3n - 41m^2n^2 - 6mn^3$

$$\begin{aligned} 7m^3n - 41m^2n^2 - 6mn^3 &= mn(7m^2 - 41mn - 6n^2) \\ &= mn(m - 6n)(7m + n) \end{aligned}$$

34. $60b^2 - 24b^4 + 4b^3$

$$\begin{aligned} 60b^2 - 24b^4 + 4b^3 &= -24b^4 + 4b^3 + 60b^2 \\ &= -4b^2(6b^2 - b - 15) \\ &= -4b^2(2b + 3)(3b - 5) \end{aligned}$$

35. (a) $18x^2 - 3xy - 10y^2$
(b) $24x - 20y - 18x^2 + 3xy + 10y^2$

$$(a) \quad 18x^2 - 3xy - 10y^2 = \underline{\underline{(3x + 2y)(6x - 5y)}}$$

$$\begin{aligned} (b) \quad 24x - 20y - 18x^2 + 3xy + 10y^2 \\ &= 4(6x - 5y) - (18x^2 - 3xy - 10y^2) \\ &= 4(6x - 5y) - (3x + 2y)(6x - 5y) \\ &= \underline{\underline{(6x - 5y)(4 - 3x - 2y)}} \end{aligned}$$

36. (a) $x^2y - 4xy^2 - 32y^3$
(b) $x^2y + 2x^2 + 16xy - 4xy^2 + 32y^2 - 32y^3$

$$\begin{aligned} (a) \quad x^2y - 4xy^2 - 32y^3 \\ &= y(x^2 - 4xy - 32y^2) \\ &= \underline{\underline{y(x + 4y)(x - 8y)}} \end{aligned}$$

$$\begin{aligned} (b) \quad x^2y + 2x^2 + 16xy - 4xy^2 + 32y^2 - 32y^3 \\ &= x^2y - 4xy^2 - 32y^3 + 2x^2 + 16xy + 32y^2 \\ &= x^2y - 4xy^2 - 32y^3 + 2(x^2 + 8xy + 16y^2) \\ &= y(x + 4y)(x - 8y) + 2[x^2 + 2(x)(4y) + (4y)^2] \\ &= y(x + 4y)(x - 8y) + 2(x + 4y)^2 \\ &= (x + 4y)[y(x - 8y) + 2(x + 4y)] \\ &= \underline{\underline{(x + 4y)(xy - 8y^2 + 2x + 8y)}} \end{aligned}$$

PRE S3 MATHS L2 – MORE ABOUT FACTORIZATION (B)

Factorize the following expressions.

37. $30m^2 - 6(12m + 9)$

$$\begin{aligned} 30m^2 - 6(12m + 9) &= 30m^2 - 72m - 54 \\ &= 6(5m^2 - 12m - 9) \\ &= \underline{\underline{6(m-3)(5m+3)}} \end{aligned}$$

38. $5n(2n - 7) - 20$

$$\begin{aligned} 5n(2n - 7) - 20 &= 10n^2 - 35n - 20 \\ &= 5(2n^2 - 7n - 4) \\ &= \underline{\underline{5(n-4)(2n+1)}} \end{aligned}$$

39. $(9p - 12)(6p + 6) - 15p$

$$\begin{aligned} (9p - 12)(6p + 6) - 15p \\ &= 54p^2 + 54p - 72p - 72 - 15p \\ &= 54p^2 - 33p - 72 \\ &= 3(18p^2 - 11p - 24) \\ &= \underline{\underline{3(2p-3)(9p+8)}} \end{aligned}$$

40. $9c - 6(c + 1)(5c - 2)$

$$\begin{aligned} 9c - 6(c + 1)(5c - 2) &= 3[3c - 2(c + 1)(5c - 2)] \\ &= 3[3c - 2(5c^2 + 5c - 2c - 2)] \\ &= 3[3c - 2(5c^2 + 3c - 2)] \\ &= 3(3c - 10c^2 - 6c + 4) \\ &= 3(-10c^2 - 3c + 4) \\ &= -3(10c^2 + 3c - 4) \\ &= \underline{\underline{-3(2c-1)(5c+4)}} \end{aligned}$$

41. $(5p + 8r)(3p - 10r) - 3p(p - 8r)$

$$\begin{aligned} (5p + 8r)(3p - 10r) - 3p(p - 8r) \\ &= 15p^2 + 24pr - 50pr - 80r^2 - 3p^2 + 24pr \\ &= 12p^2 - 2pr - 80r^2 \\ &= 2(6p^2 - pr - 40r^2) \\ &= \underline{\underline{2(2p+5r)(3p-8r)}} \end{aligned}$$

42. $(x + 3)(2x - 1) + 3(x - 9)(x + 1)$

$$\begin{aligned} (x + 3)(2x - 1) + 3(x - 9)(x + 1) \\ &= 2x^2 + 6x - x - 3 + 3(x^2 - 9x + x - 9) \\ &= 2x^2 + 5x - 3 + 3(x^2 - 8x - 9) \\ &= 2x^2 + 5x - 3 + 3x^2 - 24x - 27 \\ &= 5x^2 - 19x - 30 \\ &= \underline{\underline{(x-5)(5x+6)}} \end{aligned}$$

43. $(y + 5)^2 + 3(y + 4) - 1$

$$\begin{aligned} (y + 5)^2 + 3(y + 4) - 1 &= y^2 + 10y + 25 + 3y + 12 - 1 \\ &= y^2 + 13y + 36 \\ &= \underline{\underline{(y+4)(y+9)}} \end{aligned}$$

$$\begin{aligned} (y + 5)^2 + 3(y + 5) - 4 \\ &= [(y + 5) + 4][(y + 5) - 1] \\ &= \underline{\underline{(y + 9)(y + 4)}} \end{aligned}$$

44. $(z - 6)^2 + 2(z - 3) - 6$

$$\begin{aligned} (z - 6)^2 + 2(z - 6) \\ &= (z - 6)[(z - 6) + 2] \\ &= \underline{\underline{(z - 6)(z - 4)}} \end{aligned}$$

45. $(x + 3)^3 - 25x - 75$

$$\begin{aligned} (x + 3)^3 - 25x - 75 &= (x + 3)^3 - 25(x + 3) \\ &= (x + 3)[(x + 3)^2 - 25] \\ &= (x + 3)[(x + 3)^2 - 5^2] \\ &= (x + 3)[(x + 3) + 5][(x + 3) - 5] \\ &= \underline{\underline{(x + 3)(x + 8)(x - 2)}} \end{aligned}$$

46. $36y - 144 - (y - 4)^3$

$$\begin{aligned} 36y - 144 - (y - 4)^3 &= 36(y - 4) - (y - 4)^3 \\ &= (y - 4)[36 - (y - 4)^2] \\ &= (y - 4)[6^2 - (y - 4)^2] \\ &= (y - 4)[6 + (y - 4)][6 - (y - 4)] \\ &= (y - 4)(6 + y - 4)(6 - y + 4) \\ &= \underline{\underline{(y - 4)(y + 2)(10 - y)}} \end{aligned}$$

PRE S3 MATHS L2 - MORE ABOUT FACTORIZATION (B)

47. Factorize

- (a) $3x - 9y$,
(b) $x^2 + 2xy - 15y^2$,
(c) $x^2 + 2xy - 15y^2 - 3x + 9y$

(a) $3x - 9y = 3(x) - 3(3y)$
 $= 3(x - 3y)$
 (b) $x^2 + 2xy - 15y^2 = (x - 3y)(x + 5y)$
 (c) $x^2 + 2xy - 15y^2 - 3x + 9y$
 $= (x^2 + 2xy - 15y^2) - (3x - 9y)$
 $= (x - 3y)(x + 5y) - 3(x - 3y)$ (from (a) and (b))
 $= (x - 3y)(x + 5y - 3)$

48. Factorize

- (a) $13s + 52t$,
(b) $s^2 + 3st - 4t^2$,
(c) $s^2 + 3st - 4t^2 + 13s + 52t$.

(a) $13s + 52t = 13(s) + 13(4t)$
 $= 13(s + 4t)$
 (b) $s^2 + 3st - 4t^2 = (s - t)(s + 4t)$
 (c) $s^2 + 3st - 4t^2 + 13s + 52t$
 $= (s - t)(s + 4t) + 13(s + 4t)$ (from (a) and (b))
 $= (s + 4t)[(s - t) + 13]$
 $= (s + 4t)(s - t + 13)$

49. Factorize

- (a) $uv^2 - v^2$,
(b) $u^2 - 6u + 5$,
(c) $uv^2 - v^2 + u^2 - 6u + 5$.

(a) $uv^2 - v^2 = v^2(u) - v^2(1)$
 $= v^2(u - 1)$
 (b) $u^2 - 6u + 5 = (u - 1)(u - 5)$
 (c) $uv^2 - v^2 + u^2 - 6u + 5$
 $= v^2(u - 1) + (u - 1)(u - 5)$ (from (a) and (b))
 $= (u - 1)[v^2 + (u - 5)]$
 $= (u - 1)(v^2 + u - 5)$

50. Factorize

- (a) $-3ab^2 - 3b^2$,
(b) $a^2 + 5a + 4$,
(c) $a^2 + 5a + 4 - 3ab^2 - 3b^2$.

(a) $-3ab^2 - 3b^2 = -3b^2(a) - 3b^2(1)$
 $= -3b^2(a + 1)$
 (b) $a^2 + 5a + 4 = (a + 1)(a + 4)$
 (c) $a^2 + 5a + 4 - 3ab^2 - 3b^2$
 $= (a + 1)(a + 4) - 3b^2(a + 1)$ (from (a) and (b))
 $= (a + 1)(a + 4 - 3b^2)$

51. Factorize

- (a) $25p^2 - 16$,
(b) $5p^2q - 11pq - 12q$,
(c) $25p^2 - 16 - 5p^2q + 11pq + 12q$.

(a) $25p^2 - 16 = (5p)^2 - 4^2$
 $= (5p + 4)(5p - 4)$
 (b) $5p^2q - 11pq - 12q = q(5p^2 - 11p - 12)$
 $= q(p - 3)(5p + 4)$
 (c) $25p^2 - 16 - 5p^2q + 11pq + 12q$
 $= (25p^2 - 16) - (5p^2q - 11pq - 12q)$
 $= (5p + 4)(5p - 4) - q(p - 3)(5p + 4)$ (from (a) and (b))
 $= (5p + 4)[(5p - 4) - q(p - 3)]$
 $= (5p + 4)(5p - 4 - pq + 3q)$

52. Factorize

- (a) $14x^2 - 25x + 6$,
(b) $49x^2 - 4$,
(c) $(42x^3 - 75x^2 + 18x) - (98x^2 - 8)$.

(a) $14x^2 - 25x + 6 = (2x - 3)(7x - 2)$
 (b) $49x^2 - 4 = (7x)^2 - 2^2$
 $= (7x + 2)(7x - 2)$
 (c) $(42x^3 - 75x^2 + 18x) - (98x^2 - 8)$
 $= 3x(14x^2 - 25x + 6) - 2(49x^2 - 4)$
 $= 3x(2x - 3)(7x - 2) - 2(7x + 2)(7x - 2)$ (from (a))
 $= (7x - 2)[3x(2x - 3) - 2(7x + 2)]$
 $= (7x - 2)(6x^2 - 9x - 14x - 4)$
 $= (7x - 2)(6x^2 - 23x - 4)$
 $= (7x - 2)(x - 4)(6x + 1)$

Final Boss!

53. (a) Factorize $2x^2 - x - 10$.

(b) Hence, factorize

(i) $2(y+2)^2 - (y+2) - 10$,

(ii) $2(2z-5)^2 - 2z - 5$.

54. (a) Factorize $a^2 - 7a - 30$.

(b) Hence, factorize

(i) $(b-5)^2 - 7(b-5) - 30$,

(ii) $(3c-2)^2 - 7(3c-2) - 30$.

(a) $2x^2 - x - 10 = \underline{(x+2)(2x-5)}$

(b) (i) Let $x = y + 2$.
 $2(y+2)^2 - (y+2) - 10$
 $= 2x^2 - x - 10$
 $= (x+2)(2x-5)$ (from (a))
 $= [(y+2)+2][2(y+2)-5]$
 $= \underline{(y+4)(2y-1)}$

(ii) Let $x = 2z - 5$.
 $2(2z-5)^2 - 2z - 5$
 $= 2(2z-5)^2 - 2z + 5 - 10$
 $= 2(2z-5)^2 - (2z-5) - 10$
 $= 2x^2 - x - 10$
 $= (x+2)(2x-5)$ (from (a))
 $= [(2z-5)+2][2(2z-5)-5]$
 $= \underline{(2z-3)(4z-15)}$

(a) $a^2 - 7a - 30 = \underline{(a+3)(a-10)}$

(b) (i) Let $a = b - 5$.
 $(b-5)^2 - 7(b-5) - 30$
 $= a^2 - 7a - 30$
 $= (a+3)(a-10)$ (from (a))
 $= (b-5+3)(b-5-10)$
 $= \underline{(b-2)(b-15)}$

(ii) Let $a = 3c - 2$.
 $(3c-2)^2 - 7(3c-2) - 30$
 $= a^2 - 7a - 30$
 $= (a+3)(a-10)$ (from (a))
 $= (3c-2+3)(3c-2-10)$
 $= (3c+1)(3c-12)$
 $= (3c+1)3(c-4)$
 $= \underline{3(3c+1)(c-4)}$

55. (a) Factorize $12m^2 - 35m - 3$.

(b) Hence, factorize

$24s(4t-5)^2 - 70s(4t-5) - 6s$.

(a) $12m^2 - 35m - 3 = \underline{(m-3)(12m+1)}$

(b) Let $m = 4t - 5$.
 $24s(4t-5)^2 - 70s(4t-5) - 6s$
 $= 2s[12(4t-5)^2 - 35(4t-5) - 3]$
 $= 2s(12m^2 - 35m - 3)$
 $= 2s(m-3)(12m+1)$ (from (a))
 $= 2s[(4t-5)-3][12(4t-5)+1]$
 $= 2s(4t-8)(48t-60+1)$
 $= 2s[(4t-2)](48t-59)$
 $= \underline{8s(t-2)(48t-59)}$

56. (a) Factorize $5x^2 - 17xy + 6y^2$.

(b) Hence, factorize

$45(p-q)^2 - 51(p^2 - q^2) + 6(p+q)^2$

(a) $5x^2 - 17xy + 6y^2 = \underline{(x-3y)(5x-2y)}$

(b) Let $x = 3(p-q)$ and $y = p+q$.
 $45(p-q)^2 - 51(p^2 - q^2) + 6(p+q)^2$
 $= 5[3(p-q)]^2 - 17[3(p-q)](p+q) + 6(p+q)^2$
 $= 5x^2 - 17xy + 6y^2$
 $= (x-3y)(5x-2y)$ (from (a))
 $= [3(p-q)-3(p+q)][5[3(p-q)]-2(p+q)]$
 $= (3p-3q-3p-3q)(15p-15q-2p-2q)$
 $= \underline{-6q(13p-17q)}$

57. (a) Factorize $p^2 - 13p + 36$.

(b) Hence, factorize $q^4 - 13q^2 + 36$.

(a) $p^2 - 13p + 36 = \underline{(p-4)(p-9)}$

(b) Let $p = q^2$.
 $q^4 - 13q^2 + 36$
 $= (q^2)^2 - 13q^2 + 36$
 $= p^2 - 13p + 36$
 $= (p-4)(p-9)$ (from (a))
 $= (q^2-4)(q^2-9)$
 $= (q^2-2^2)(q^2-3^2)$
 $= \underline{(q+2)(q-2)(q+3)(q-3)}$

58. (a) Factorize $b^2 - 14b + 40$.

(b) Hence, factorize $(a^2 - 3a)^2 - 14(a^2 - 3a) + 40$.

(a) $b^2 - 14b + 40 = \underline{(b-4)(b-10)}$

(b) Let $b = a^2 - 3a$.
 $(a^2 - 3a)^2 - 14(a^2 - 3a) + 40$
 $= b^2 - 14b + 40$
 $= (b-4)(b-10)$ (from (a))
 $= (a^2 - 3a - 4)(a^2 - 3a - 10)$
 $= \underline{(a+1)(a-4)(a+2)(a-5)}$

59. (a) Factorize $5a^2 - 31a + 6$.
 (b) Hence, factorize $5b^3 - 31b^2 + 51b - 9$.

(a) $5a^2 - 31a + 6 = \underline{(a-6)(5a-1)}$

(b)
$$\begin{aligned} 5b^3 - 31b^2 + 51b - 9 &= 5b^3 - 31b^2 + 6b + 45b - 9 \\ &= b(5b^2 - 31b + 6) + 45b - 9 \quad (\text{from (a)}) \\ &= b(b-6)(5b-1) + 9(5b-1) \\ &= (5b-1)[b(b-6)+9] \\ &= (5b-1)(b^2-6b+9) \\ &= (5b-1)[b^2-2(b)(3)+3^2] \\ &= \underline{(5b-1)(b-3)^2} \end{aligned}$$

60. (a) Factorize $20y^2 - 49y + 30$.
 (b) Hence, factorize $4y^3 + 15y^2 - 49y + 30$.

(a) $20y^2 - 49y + 30 = \underline{(4y-5)(5y-6)}$

(b)
$$\begin{aligned} 4y^3 + 15y^2 - 49y + 30 &= (4y^3 - 5y^2) + (20y^2 - 49y + 30) \\ &= y^2(4y-5) + (4y-5)(5y-6) \quad (\text{from (a)}) \\ &= (4y-5)[y^2 + (5y-6)] \\ &= (4y-5)(y^2 + 5y - 6) \\ &= \underline{(4y-5)(y-1)(y+6)} \end{aligned}$$

61. (a) Factorize $6x^2 - 7x - 10$.
 (b) Hence, simplify $\frac{6x^2 - 7x - 10}{4x - 8} \times \frac{2}{6x + 5}$.

(a) $6x^2 - 7x - 10 = \underline{(x-2)(6x+5)}$

(b)
$$\begin{aligned} \frac{6x^2 - 7x - 10}{4x - 8} \times \frac{2}{6x + 5} &= \frac{(x-2)(6x+5)}{4x-8} \times \frac{2}{6x+5} \quad (\text{from (a)}) \\ &= \frac{(x-2)(6x+5)}{4(x-2)} \times \frac{2}{6x+5} \\ &= \underline{\underline{\frac{1}{2}}} \end{aligned}$$

62. (a) Factorize
 (i) $9x^2 - 16y^2$,
 (ii) $15x^2 - xy - 28y^2$.
 (b) Hence, simplify $\frac{15x^2 - xy - 28y^2}{45x - 63y} \div \frac{9x^2 - 16y^2}{15x - 6y}$.

(a) (i) $9x^2 - 16y^2 = (3x)^2 - (4y)^2$
 $= \underline{(3x+4y)(3x-4y)}$

(ii) $15x^2 - xy - 28y^2 = (3x+4y)(5x-7y)$

(b)
$$\begin{aligned} \frac{15x^2 - xy - 28y^2}{45x - 63y} \div \frac{9x^2 - 16y^2}{15x - 6y} &= \frac{(3x+4y)(5x-7y)}{45x-63y} \div \frac{(3x+4y)(3x-4y)}{15x-6y} \quad (\text{from (a)}) \\ &= \frac{(3x+4y)(5x-7y)}{9(5x-7y)} \times \frac{3(5x-2y)}{(3x+4y)(3x-4y)} \\ &= \frac{1}{3} \times \frac{5x-2y}{3x-4y} \\ &= \underline{\underline{\frac{5x-2y}{3(3x-4y)}}} \end{aligned}$$

63. (a) Factorize $x^2 + 5x - 24$.
 (b) Hence, simplify $\frac{x}{x^2 + 5x - 24} - \frac{1}{x+8}$.

(a) $x^2 + 5x - 24 = \underline{(x-3)(x+8)}$

(b)
$$\begin{aligned} \frac{x}{x^2 + 5x - 24} - \frac{1}{x+8} &= \frac{x}{(x-3)(x+8)} - \frac{1}{x+8} \quad (\text{from (a)}) \\ &= \frac{x}{(x-3)(x+8)} - \frac{x-3}{(x-3)(x+8)} \\ &= \frac{x - (x-3)}{(x-3)(x+8)} \\ &= \underline{\underline{\frac{3}{(x-3)(x+8)}}} \end{aligned}$$

64. (a) Factorize $3x^2 - 13x - 10$.
 (b) Hence, simplify $\frac{1}{x-5} - \frac{2x+7}{3x^2 - 13x - 10}$.

(a) $3x^2 - 13x - 10 = \underline{(x-5)(3x+2)}$

(b)
$$\begin{aligned} \frac{1}{x-5} - \frac{2x+7}{3x^2 - 13x - 10} &= \frac{1}{x-5} - \frac{2x+7}{(x-5)(3x+2)} \quad (\text{from (a)}) \\ &= \frac{3x+2}{(x-5)(3x+2)} - \frac{2x+7}{(x-5)(3x+2)} \\ &= \frac{3x+2 - (2x+7)}{(x-5)(3x+2)} \\ &= \frac{3x+2-2x-7}{(x-5)(3x+2)} \\ &= \frac{x-5}{(x-5)(3x+2)} \\ &= \underline{\underline{\frac{1}{3x+2}}} \end{aligned}$$